

Cybersecurity Incident Report:

Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.

The UDP protocol reveals that:

The browser sent UDP packets to the DNS server to request the IP address for yummyrecipesforme.com. The DNS requests were repeatedly unsuccessful because the DNS service could not be reached.

This is based on the results of the network analysis, which show that the ICMP echo reply returned the error message:

udp port 53 unreachable

The port noted in the error message is used for:

Port 53 is used by the DNS (Domain Name System) service to translate domain names into IP addresses.

The most likely issue is:

The DNS server was unable to receive or process DNS requests on port 53. As a result, the browser could not obtain the IP address of the website, preventing users from accessing the site.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.

Time incident occurred:

The first error was observed at approximately 13:24:36, shortly after the initial DNS request at 13:24:32.

Explain how the IT team became aware of the incident:

Multiple users reported that they could not access the website www.yummyrecipesforme.com and received the message "destination port unreachable." The IT team reproduced the issue and began investigating with the tcpdump network protocol analyzer.

Explain the actions taken by the IT department to investigate the incident:

The IT team captured and analyzed network traffic using tcpdump while attempting to

access the website. They reviewed DNS requests and the corresponding responses from the DNS server.

Note key findings of the IT department's investigation (i.e., details related to the port affected, DNS server, etc.):

- DNS requests were sent using the UDP protocol.
- The requests were directed to the DNS server at IP address 203.0.113.2.
- The DNS server returned ICMP error messages.
- The error message repeatedly stated "udp port 53 unreachable."
- Multiple DNS requests failed, preventing resolution of the website's domain name.

Note a likely cause of the incident:

A DNS service failure on port 53 is the most likely cause. The DNS server may have been misconfigured, offline, blocked by a firewall, or not running a DNS service on port 53, preventing successful DNS resolution.